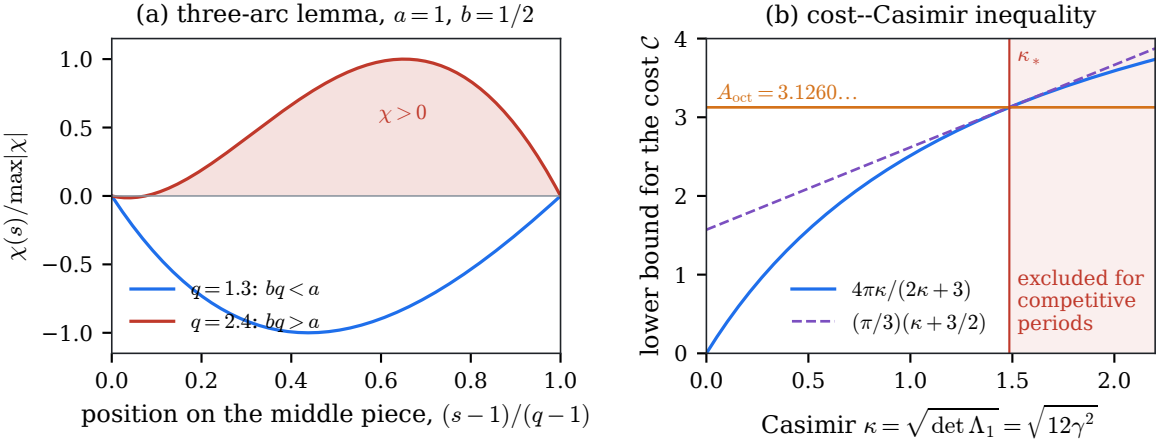
(a) three-arc lemma, $a=1$, $b=1/2$
$\chi(s)/\max|\chi|$
1.0
0.5
0.0
−0.5
−1.0
$\chi>0$
$q=1.3$: $bq<a$
$q=2.4$: $bq>a$
0.0
0.2
0.4
0.6
0.8
1.0
position on the middle piece, $(s-1)/(q-1)$
(b) cost--Casimir inequality
lower bound for the cost C
4
3
2
1
0
$A_{\rm oct}=3.1260\ldots$
κ_*
$4\pi\kappa/(2\kappa+3)$
$(\pi/3)(\kappa+3/2)$
excluded for competitive periods
0.0
0.5
1.0
1.5
2.0
Casimir $\kappa=\sqrt{\det\Lambda_1}=\sqrt{12\gamma^2}$